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Ultrafast laser micro-welding of glass to SiC: Microstructure, mechanical and impermeable properties

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Abstract

Ultrafast laser micro-welding was used for the high-efficient and high-quality joining of aluminosilicate glass (AS) and SiC, whose reliable packaging was of importance in Micro-Electro-Mechanical Systems (MEMS). By using the orthogonal test method, the process parameters were optimized, which were single pulse energy of 30 μJ, scanning speed of 10 mm/s, and the scanning pattern of filling grid inside circle. Under these conditions, the joint achieved a maximum shear strength of 33.1 MPa. Microstructural analysis revealed an evident elemental mixture layer with a thickness of 1.4 μm at the interface, indicating strong material intermixing and elemental diffusion under ultrafast laser irradiation. In addition, the joint permeability was assessed by waterproofing evaluations, revealing that the joints exhibit outstanding impermeable properties, which met the IPX7 waterproof standard (immersion in 1 m of water for 30 min) for enclosure packaging in practical applications. Furthermore, the temperature field simulation results showed that the temperature generated by the laser irradiation during the welding process exceeded the softening temperature and melting point of the base material. The joint formation mechanism was also discussed. This work provides theoretical guidance for the ultrafast laser impermeability of transparent materials with SiC, holding significant potential in applications like semiconductor packaging and MEMS fabrication.

Previous

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Keywords

Ultrafast laser; Micro-welding; Aluminosilicate glass/SiC; Mechanical property; Impermeability

1. Introduction

Micro-Electro-Mechanical Systems (MEMS) are composed of various components, including sensors, actuators, passive components, power management circuits, as well as analog and digital integrated circuits. MEMS convert environmental process parameters into electrical signals and vice versa [1,2]. MEMS devices contain movable fragile parts. Without proper protection, the movable components were easily damaged [3]. Additionally, for the key reliability indicators such as corrosion resistance, moisture resistance, thermal cycle tolerance, and mechanical stability faced by the device during operation [4,5]. Therefore, MEMS packaging also needs to be customized and strictly verified for the material system (airtight packaging materials and biocompatible media) and integrated process (wafer-level packaging and three-dimensional heterogeneous integration), based on the specific application scenarios [6,7].

Silicon (Si), as the first-generation semiconductor material, has dominated MEMS fabrication for decades. However, Si-based devices exhibit limited reliability under harsh operating conditions such as high temperatures (>300 °C), radiation exposure, and corrosive environments [8,9]. The emergence of third-generation semiconductors, such as silicon carbide (SiC), is able to solve these limitations. In terms of SiC, it has a wide band gap (3.3 eV), a high breakdown electric field (3 MV/cm), excellent thermal conductivity (490 W/(m·K)), and excellent mechanical hardness (28 GPa) [[10], [11], [12]]. Aside from the semiconductor materials, glass, as a protective cover plate, is also a kind of important component in MEMS manufacturing, *e.g.*, solar cells, thermal sensors, and microfluidic chips [13]. That is because glass has the advantages of excellent light transmittance, high insulation, and superior chemical stability [14,15]. Therefore, the integration of glass and SiC offers a promising pathway for next-generation MEMS devices operating in extreme environments [16,17]. Nevertheless, achieving robust heterojunctions between these materials remains a challenge due to their inherent physicochemical disparities (*e.g.*, large differences in melting points and mismatch in Young's moduli, *etc.*).

The traditional methods for the joining of glass and semiconductor include adhesive bonding [18], anodic bonding [19], solid phase bonding [20], and eutectic bonding [21]. However, drawbacks such as contamination, aging, low accuracy, and non-selectability are exhibited by these methods [19,[22], [23], [24]]. Therefore, a non-polluting, high-precision and high-efficient bonding technique is urgently needed.

Ultrafast laser are characterized by pulse widths on the order of tens of femtoseconds to tens of picoseconds [25]. Such ultrashort pulse widths limit heat diffusion into the surrounding regions, which significantly reduces the heat-affected zone (HAZ) and enables ultrahigh precision micro- and nanofabrication for various materials [26]. Owing to the ultrashort pulse width, ultrafast laser generate extremely high peak power densities (>10 PW/cm^2), which can induce nonlinear multiphoton absorption within transparent hard and brittle (THB) materials [[27], [28], [29]]. When focused at or near the THB/SiC interface, the laser achieves sufficient intensity within the focal volume to induce nonlinear absorption in the material, leading to localized melting at the interface and rapid filling of the original gaps [30,31]. The liquid then solidifies rapidly, eventually forming a micro-weld at the interface [32]. Due to the highly localized nature of ultrafast laser, the feature size of the weld region can reach the micrometer level, achieving micrometer-scale welding precision. Fused quartz glass and SiC were welded using the femtosecond laser micro-welding technique by Zhang et al. [33]. The joint peak shear strength of 15.1 MPa was achieved. Alexander et al. [34] investigated the relationship between the weld dimensions and geometry and process parameters, including scanning speed, repetition rate, and pulse energy, in ultrafast laser welding of glass/glass and glass/Si. Zhang et al. [35] successfully connected glass and monocrystalline silicon using ultrafast laser welding. By optimization of welding parameters by orthogonal experiments, a shear strength of 12 MPa was ultimately achieved. Despite extensive research efforts, the following issues remain unresolved: (i) systematic correlations between laser processing parameters (*e.g.*, power density, scanning pattern) and joint properties (*e.g.*, mechanical property, permeability) have not been established systematically, and (ii) the joint bonding mechanism remains unverified.

In the present study, the process parameters for ultrafast laser micro-welding of aluminosilicate glass (AS) and silicon carbide (SiC) were systematically optimized. With the optimized parameters (single pulse energy of 30 μJ, scanning speed of 10 mm/s, scanning pattern of filling grid inside circle), the peak shear strength of 33.1 MPa was achieved. According to the COMSOL-based simulations, the temperature field evolution during the micro-welding process was also revealed, which was then used for the discussion of the joint formation mechanism.

2. Experimental procedure

AS sheets with dimensions of 15 mm × 15 mm × 0.7 mm and SiC sheets with dimensions of 15 mm × 15 mm × 0.5 mm were used as base materials (BMs). Fig. 1a and (c) present optical images of the initial surface states of BMs. Fig. 1b and (d) show the surface roughness of BMs using a laser confocal microscope (LSCM, OLS5000), with color mapping used to characterize changes in surface height. The average surface roughness of AS was 0.965 μm, and the average surface roughness of SiC was 0.624 μm. The chemical compositions of the BMs are shown in Table 1, where the AS is mainly composed of SiO_2, Al_2O_3, and B_2O_3.

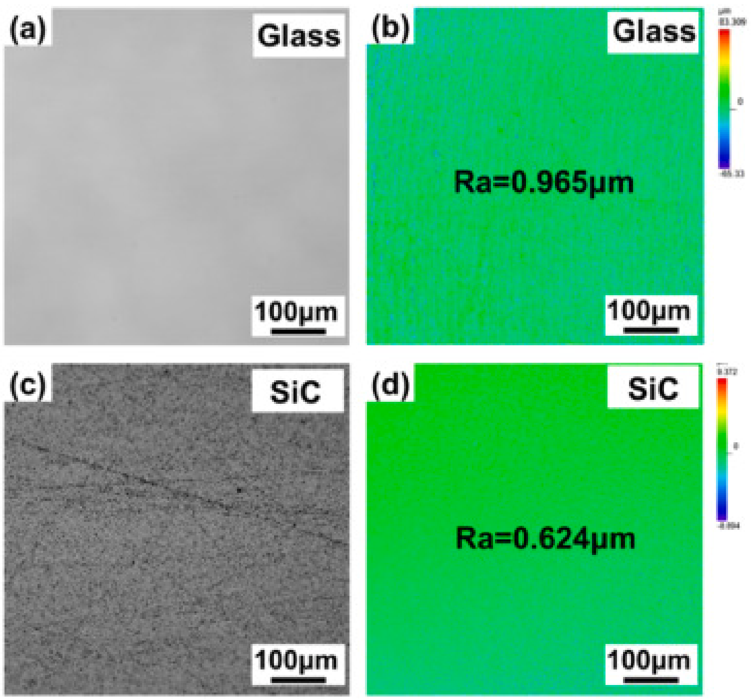

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Fig. 1. Confocal microscope image of (a–b) AS and (c–d) SiC, showing the surface roughness of the materials.

Table 1. Chemical compositions of AS and SiC (at%).

Materials	Elements				
	C	Si	O	Al	Others
AS	/	21.9	63.2	8.1	6.8
SiC	54.8	45	/	/	0.2

Ultrafast laser micro-welding of AS and SiC demands strict interfacial gap control, that is, smaller than a quarter of the minimum wavelength in the visible spectrum, to attain optical contact [36]. Consequently, before welding, the AS and SiC BMs were ultrasonically cleaned in alcohol for 5 min to remove particulate and organic contaminants. During the welding process, a specific fixture was used to apply a certain amount of pressure to the material to achieve optical contact, generating Newton's rings, as shown in Fig. 2.

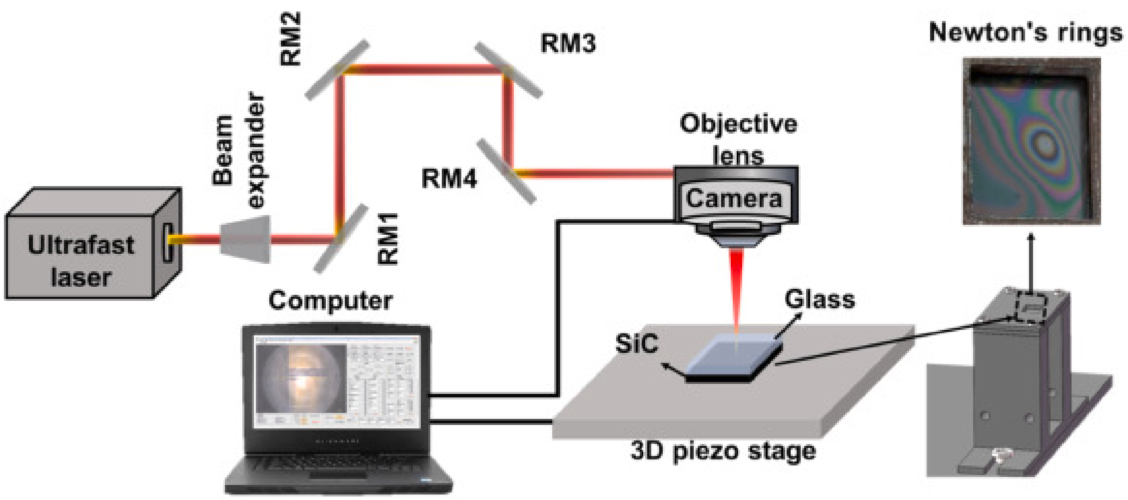

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Fig. 2. Illustration of the experimental setup for ultrafast laser micro-welding.

An ultrafast laser source, Femto YL-40 (Wuhan Anyang Laser Technology Co.), was used for the laser micro-welding experiments with a wavelength of 1030nm, a repetition frequency of 0.025–5MHz, and a focused laser spot diameter of 40μm. After preliminary trials, the process parameters were chosen as single pulse energies of 10–30μJ (peak fluence: 1.59–4.77J/cm^{-2}), repetition frequency of 200kHz, pulse width of 300fs, scanning speeds of 1–10mm/s (linear Energy Density: 0.2–6 J/mm), and scanning line spacing of 100μm. The scanning modes include: grid scanning, concentric circles, and filling grid inside circle. To provide a quantitative basis for the energy deposition [[37], [38], [39]], several Figures of Merit (FOM) were calculated. The Linear Energy Density (E_{line})and Peak Fluence (F_{peak}) were defined as:

$$E_{line} = \frac{E_P \cdot f_{rep}}{v} \tag{1}$$

$$F_{peak} = \frac{2E_P}{\pi\omega_0^2} \tag{2}$$

where E_P was the single pulse energies (μJ), f_{rep} was the repetition frequency (kHz), v was the scanning speed (mm/s), ω_0 was the Gaussian beam radius.

Before welding, the AS and SiC were assembled in an overlapping configuration and clamped using a self-designed fixture. The fixture contributed to regulating assembly forces and gaps and played an important role in restricting the escape of molten material and plasma. During the welding process, the laser beam was focused vertically on the contact interface of the two materials from the upper side of the AS. The schematic diagram of the equipment setup is shown in Fig. 2.

Orthogonal testing is a statistical approach that enables the structured study of multiple factors across various levels within a limited number of experimental trials. By selecting a suitable number of representative tests from a large number of test data, which have evenly dispersed, neat, and comparable characteristics. The results of orthogonal experiments, following calculation and analysis, can effectively shorten testing time, reduce the testing period, and rapidly identify the optimal program. In the present study, the experiments were based on an orthogonal array experimental design where the following three variables were analyzed: pulse energy (factor A), scan speed (factor B), and scan pattern (factor C). Table 2 presents the process parameters selected and their designated levels.

Table 2. Parameters influencing the shear strength of the joint.

Levels	Factors		
	A: Single pulse energy (μJ)	**B: Scanning speed (mm/s)**	**C: Scanning pattern**
1	10	1	grids
2	20	5	Filling grid inside circle
3	30	10	concentric circles

The experiment was conducted using an L9 orthogonal array. Table 3 enumerates the parameters for laser welding within the orthogonal test. To ensure experimental reliability, each experimental group was repeated five times.

Table 3. Orthogonal array of ultrafast laser micro-welding parameters.

No.	A	B	C
1	10	1	Grids
2	10	5	Filling grid inside circle
3	10	10	concentric circles
4	20	1	Filling grid inside circle
5	20	5	concentric circles
6	20	10	Grids
7	30	1	concentric circles
8	30	5	Grids
9	30	10	Filling grid inside circle

In orthogonal experiments, mechanical properties served as key indicators to evaluate micro-weld performance across different process parameters. Comparing shear strength across experimental groups identified the factors most affecting material mechanical behavior, facilitating optimization of design and production processes.

To investigate interface morphology, metallurgical specimens were ground and polished, and joint macro-morphology was examined using ultra-deep field microscopy (UFM). In addition, scanning electron microscope (SEM, ZEISS Gemini SEM 300) and energy dispersive spectroscopy (EDS, AMETEK) characterized interfacial microstructure and elemental distribution. The crystal structure of the samples before and after the laser micro-welding was determined by X-ray diffraction (XRD, Ultima IV, Rigaku, Japan). Shear strength testing with custom shear fixtures mounted on a universal mechanical testing machine (SFL25AG) characterized the mechanical properties of the joints. Pressure was exerted at a constant speed of 0.5 mm/s, with a minimum of three samples tested per parameter set. The universal testing machine is shown in Fig. 3a. The average of each process parameter was recorded as F. The weld shear strength σ was calculated using equation (1).

$$\sigma = \frac{F}{S} \tag{1}$$

Where S was the welding area (as shown in Fig. 3b).

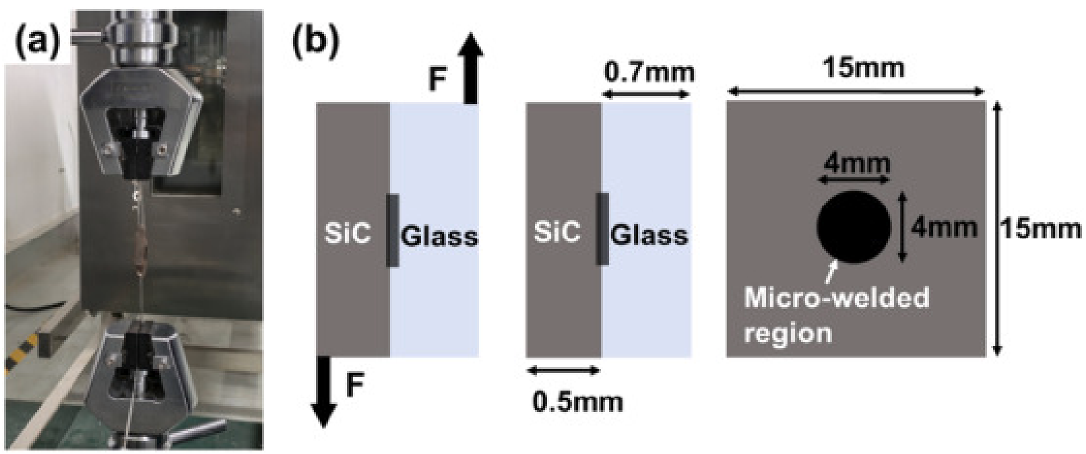

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Fig. 3. (a) Universal testing machine; (b) schematic diagram of the shear test.

The numerical simulation analysis was developed using COMSOL Multiphysics 6.3. The joint formation mechanism was analyzed by observing the temperature change. In the COMSOL temperature field simulation, part of the literature uses the two-temperature model (TTM) for simulation; the theoretical equations of this model involve the lattice temperature and the electron temperature. In addition, the calculation cost of this model was higher. Preliminary studies have shown little difference in the computational results between the direct geometry model and the dual-temperature model [40,41]. To reduce the amount of computation, therefore, in this model, the material was homogeneous, and there was uniform heat transfer, and metallurgical phenomena were not considered. A moving Gaussian pulsed heat source was used to represent the laser, as expressed in Equations (2) and (3).

$$Q = (1 - R) \times an1\left(t\left[\frac{1}{s}\right]\right) \times \frac{A \times 2 \times p}{\tau_p \times \pi \times r^2} exp\left[-\frac{(x - v \times t)^2 + y^2 + z^2}{r^2}\right] \tag{2}$$

$$an1 = \left\{\begin{matrix} 0, \tau_p & 1 \\ (\Delta_\tau + \tau_p), \Delta_\tau & 0 \\ \Delta_\tau, (\Delta_\tau + \tau_p) & 1 \end{matrix}\right\}, an1 = (t \geq 0) \&\& (t \leq \tau_p) \tag{3}$$

where, an1(t[1/s]) was a segmented function to represent the action and relaxation of the ultrashort laser, *p* was the laser pulse energy (J), τ_p was the pulse width (fs), *r* was the radius of the laser spot (μm), Δ_τ was the pulse period, *A* was the absorption coefficient of the material, *R* was the reflection coefficient of the material, *x*, *y* and *z* were the instantaneous coordinates of the spot center in the cartesian coordinate system, and *v* was the moving speed of the laser (mm/s).

To study the temperature response of the materials during the laser micro-welding process, a pulse period of 5 μs, a pulse width of 300fs, a repetition rate of 200kHz, and a laser welding speed of 10mm/s were applied for the numerical simulation.

Temperature probes were placed at positions (0, 0, 1) and (150, 0, 1) on the upper surface of the AS side, and at positions (0, 0, −1) and (150, 0, −1) on the upper surface of the SiC side, as shown in Fig. 4. The purpose was to monitor the temperature changes of the AS and SiC as the ultrashort pulsed laser heat source moved at the interface.

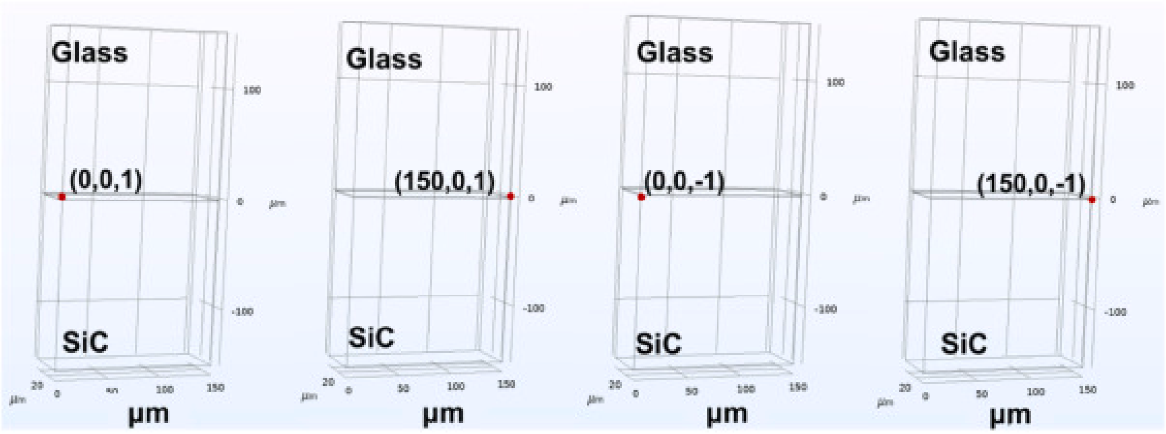

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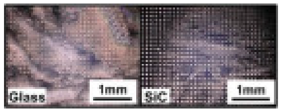
Fig. 4. Temperature probe at positions: (a) (0, 0, 1) and (b) (150, 0, 0) on the upper surface of the AS side and at positions (c) (0, 0, −1) and (d) (150, 0, −1) on the upper surface of the SiC side.

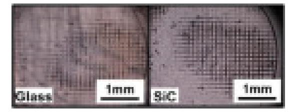
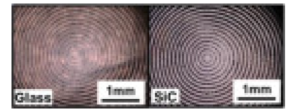
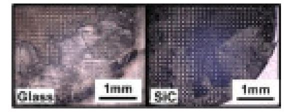
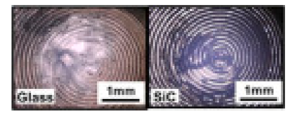
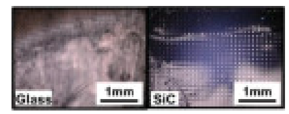
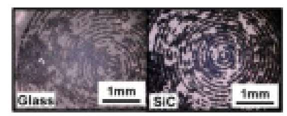
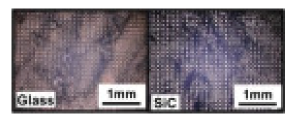
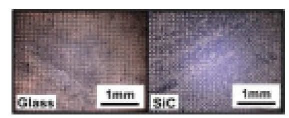
3. Results and discussions

3.1. Process parameter optimization

Table 4 shows shear strength and fracture surface morphology of the joints obtained with the orthogonal test design in Table 3. It was seen that the shear strength ranged between 2.7 MPa and 33.1 MPa, where the maximum shear strength was achieved at No.9. Macroscopic morphological observation revealed that joint fracture mainly occurred at the interface.

Table 4. Analysis of the results from the orthogonal test.

No.	A	B	C	Shear strength (MPa)	Fracture surface morphology
1	10	1	grids	18.1	

No.	A	B	C	Shear strength (MPa)	Fracture surface morphology
2	10	5	Filling grid inside circle	18.7	
3	10	10	concentric circles	9.0	
4	20	1	Filling grid inside circle	22.0	
5	20	5	concentric circles	13.9	
6	20	10	grids	20.3	
7	30	1	concentric circles	2.7	
8	30	5	grids	24.4	
9	30	10	Filling grid inside circle	33.1	

The extreme differences between the three process parameters and the shear strength were calculated as shown in Table 5. The range value (R) for each factor was the difference between the maximum and minimum value of the three factors. The R directly reflected the relative extent to which each factor influenced the variation in the test. The greater the R of a factor, the more significant its contribution to experimental outcomes [42]. According to Table 5, the R for the three factors were 14.35, 19.60, and 48.18. Based on the results, the three factors contributed to the mechanical properties of the joints in the following order of decreasing influence: scanning pattern, scanning speed, and single pulse energy. For each factor, the maximum value recorded at its respective level was selected. It was able to determine the optimal conditions A3B3C2 (No.9), with parameters of a single pulse energy of 30 μJ, a scanning speed of 10 mm/s, and a scanning pattern of a filling grid inside a circle.

Table 5. Three-factor extreme difference analysis of the orthogonal test design.

Items	**A**	**B**	**C**
Kw1	45.87	42.89	62.80
Kw2	56.30	57.01	73.88
Kw3	60.21	62.49	25.70
Range	14.35	19.60	48.18
Influence	Scan pattern > Scan speed > Pulse energy		
Optimized	3	3	2

3.2. Macrostructure

Fig. 5 illustrates the macrostructure of the micro-weld with optimal process parameters. Notably, the micro-welded peripheral region also observed Newton's rings, which were characterized by colored fringes (Fig. 5a). Newton's rings are defined as concentric interference fringes resulting from the reflection of light between two surfaces, with an air gap that increases radially from zero at the point of contact [43]. The results indicated that solid bonds were formed between AS and SiC after ultrafast laser irradiation. In addition, no edge fractures and residues were observed around the interface, and the thickness of the micro-welds was uniform with neat edges, which meant that the ultrafast laser had very limited damage to the base materials. Fig. 5b shows the macroscopic cross-sectional morphology of the AS and SiC micro-welds. A distinct dark gray reaction zone was observed at the interface, and the distance between each weld point was found to correspond to the 100 μm line spacing. Additionally, an obvious modification was seen at the AS side. The reasons for the AS modified zone were likely caused by the plasma evolution, thermal shock, and thermal expansion of the material at the laser focus during the ultrafast laser selective welding process [44,45].

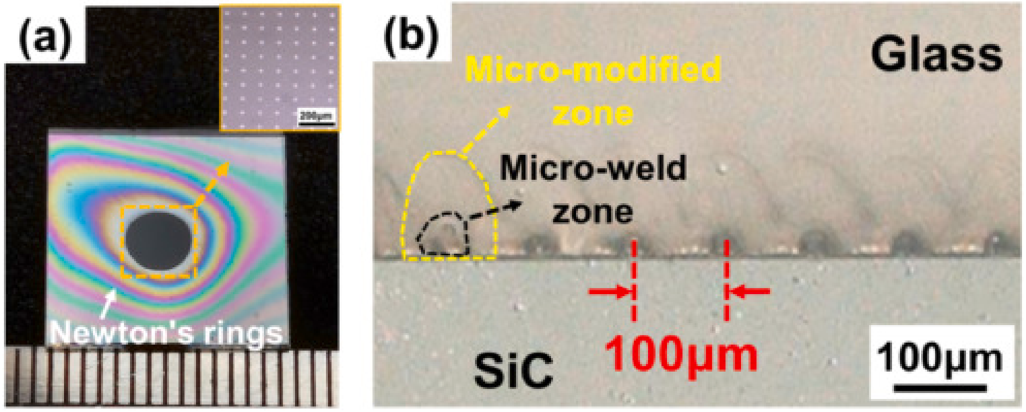


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Fig. 5. Morphology of ultrafast laser AS/SiC micro-weld: (a) macrostructure; (b) cross-sectional view.

3.3. Microstructure

Fig. 6 shows the SEM and EDS analysis results of the cross sections of the micro-welds. Fig. 6a shows that periodic bonded interfaces were formed between AS and SiC after ultrafast laser irradiation, leaving some original gaps at the interfacial region. At a higher magnification, a tortuous interface was observed between AS and SiC (Fig. 6b). It was noted that the uneven interface was likely to provide extra micro-mechanical interlocking at the interface [46]. In addition, a mixture layer transition region was observed. It was probably caused by the materials intermixing and laser-material interaction at the interface, which will be discussed later. According to the EDS elemental mapping analysis in Fig. 6(c-f), it was observed that the O element and the Al element were enriched at the interface. This phenomenon indicates the occurrence of microscopic material mixing and interdiffusion in the vicinity of the bonding interface during ultrafast laser irradiation. Moreover, the EDS line scanning results verified the evident elemental transition layers, with thicknesses approximately 1.4μm (Fig. 6g). It was believed that the mixed material near the interface acts as a buffer layer, which could effectively alleviate the mismatch between the thermal expansion coefficients and melting points of AS and SiC. [33]. The synergistic influence of tortuous interface and elemental transition layer thereby facilitated robust bonding between AS and SiC.

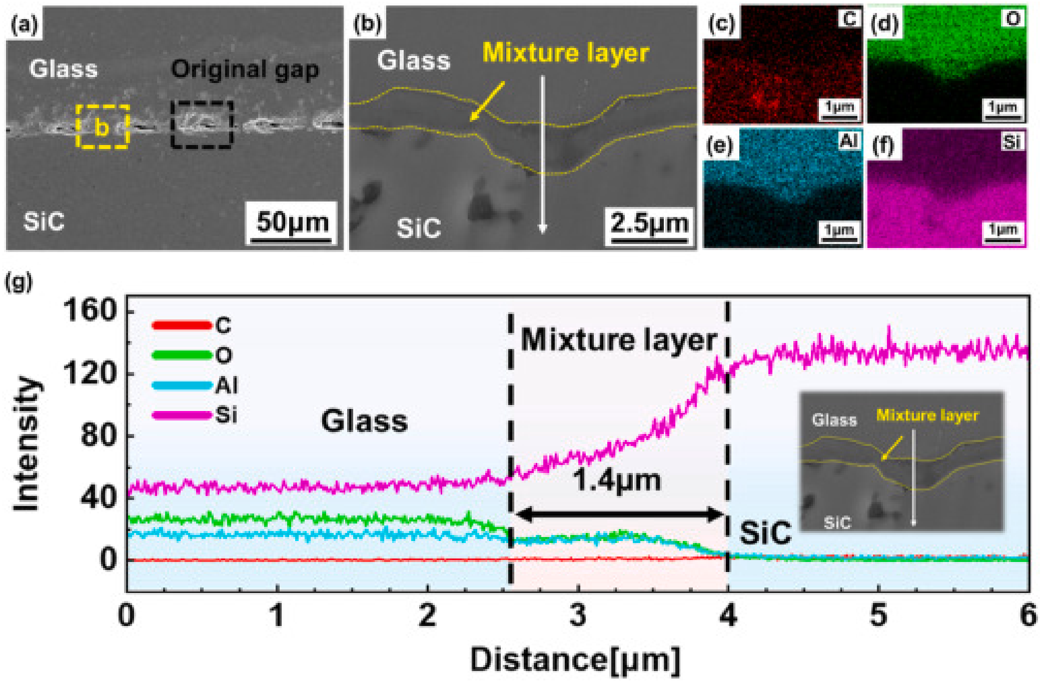

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Fig. 6. (a–b) The typical microstructure at the interfacial region of AS/SiC micro-weld; (c)–(f) EDS mapping analysis in (b); (g) EDS line scanning in (b).

3.4. Fractography

Fig. 7, Fig. 8 show the joint fracture morphology at the AS side and the SiC side at optimum parameters, respectively. It was noted that the failure occurred at the interfacial region in the micro-welds (Table 4).

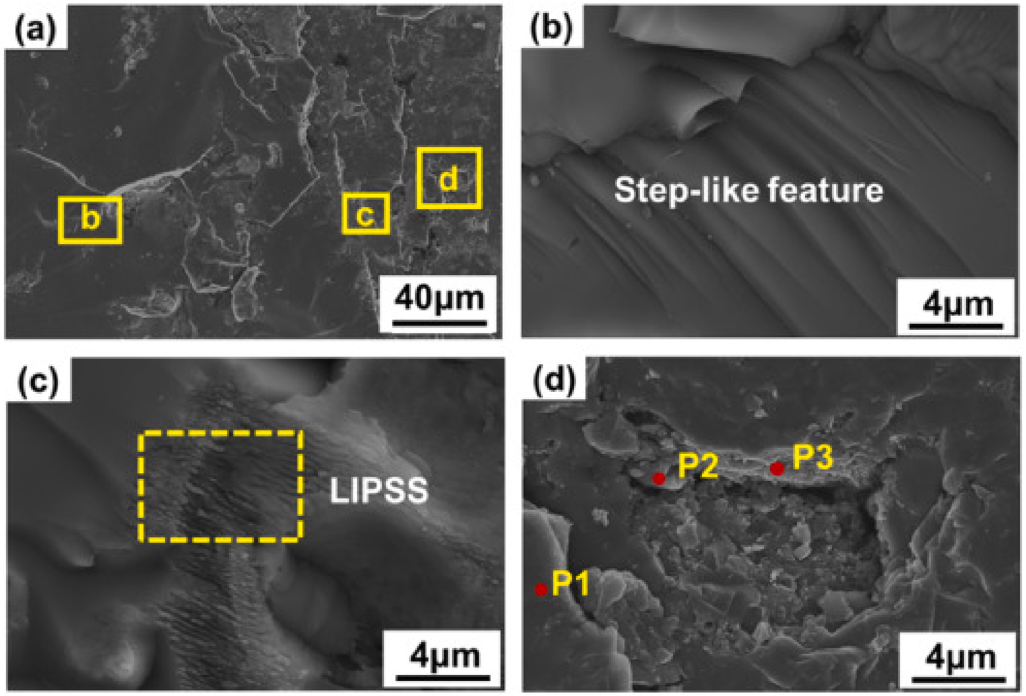

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Fig. 7. The typical fracture surface morphology at the AS side: (a) SEM images of fracture surfaces of AS welded region; (b–d) the magnified image of the marked zone in (a).

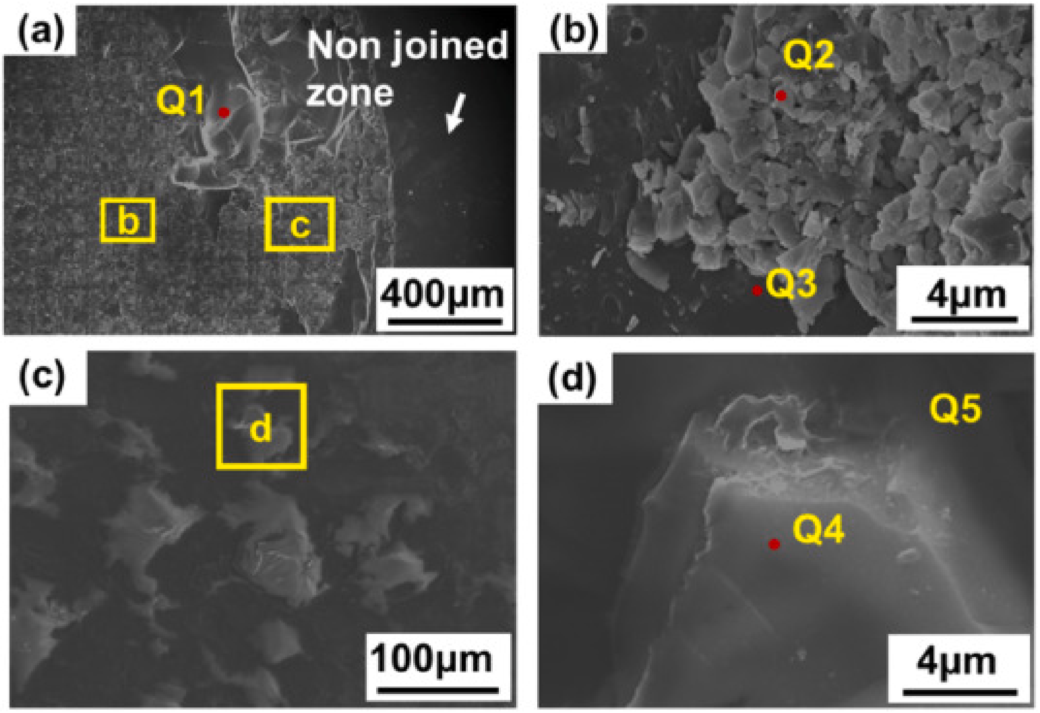

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Fig. 8. The typical fracture surface morphology at SiC side: (a) the microstructure of the fracture surface occurred at the interface; (b) the magnified image of the marked zone b in (a); (c) the magnified image of the marked zone in (a); (d) the magnified image of the marked zone in (c).

Fig. 7 presents the fracture surface morphology at the SiC side. A relatively flat fracture surface was observed, indicating a brittle failure, owing to the brittle nature of the AS and SiC, as shown in Fig. 7a. Some step-like features were evident on the fracture surface, which was also characteristic of brittle fracture (Fig. 7b). Some ripple-like structures were seen on the AS fracture side (Fig. 7c). The ripple-like morphology, a characteristic ultrafast laser-induced structure, was termed as laser-induced periodic surface structure (LIPSS) [47]. This feature contributed to the formation of a sound joint by ultrafast laser micro-welding. Additionally, Fig. 7d presents a high-magnification view of the fracture surface. A number of pit-like structures were observed. Within the pit-like structure, some brittle fragments were found, which were mainly ascribed to the melting and solidification of

AS near the interface. According to the EDS analysis (Table 6), the contents of elements Si and C in P1 were significantly higher than those of the AS base material, indicating that the region was possibly attached to SiC, and the fracture occurred on the SiC side. The content of Si and C elements at P2 was close to that of the SiC base material, and the SiC was attached to the AS side. C content at point P3 was significantly higher than that in the SiC base material, indicating that C enrichment occurred in this region. The reason for this phenomenon was the high temperature caused by the ultrafast laser irradiation, leading to the decomposition of SiC, with free C enriched in this region [48].

Table 6. EDS elemental composition of the micro-zones marked in Fig. 7(d) (at.%).

Points	Elements			
	C	**O**	**Al**	**Si**
P1	14.16	43.64	8.82	33.38
P2	54.13	2.28	0.01	43.58
P3	71.84	1.37	0.03	26.76

Fig. 8 presents the fracture surface morphology at the SiC side. A significant amount of bulged materials was evident across the surface (Fig. 8a). EDS analysis (Table 7) identified the smooth structures at point Q1 as the AS BM, confirming that fracture partially occurred within the AS BM. In the region with rough structures, numerous fragments were observed on the fracture surface (Fig. 8b-d). According to the EDS analysis of points Q2 and Q4, a high O content (approximately 54–56at.%) was detected, indicating the presence of AS residuals on the fracture surface (Table 7). However, the EDS analysis on Q3 and Q5 confirmed the existence of mixed zones of AS and SiC.

Table 7. EDS elemental composition of the micro-zones marked in Fig. 8(a–d) (at.%).

Points	Elements			
	C	**O**	**Al**	**Si**
Q1	16.04	48.00	6.40	29.56

Points	Elements			
	C	O	Al	Si
Q2	14.74	56.74	7.18	21.34
Q3	12.14	17.94	13.53	56.39
Q4	20.24	54.63	6.31	18.82
Q5	53.49	8.12	0.36	38.03

Based on the EDS analysis of both fracture surfaces (Table 6, Table 7), it was revealed that the C, O, Al, and Si elements were distributed on both sides of the fracture surfaces owing to material intermixing and interdiffusion. It further demonstrated that the ultrafast laser process effectively promoted material interaction and final bonding between AS and SiC. Based on the fractography analysis, the fracture was identified as a mixed mode, occurring at both the substrates and the interfacial region (mixed zone).

To further identify the phase composition in the micro-weld, XRD analysis was conducted on the SiC side of the fracture surfaces (Fig. 9). As shown in Fig. 9a, before micro-welding, it showed that the 6H-SiC phase was the main phase in SiC. However, after micro-welding, additional phases of fused silica (SiO_2) and aluminum oxide (Al_2O_3) were identified, which probably originated from the AS side. The results indicated that materials intermixing and elemental diffusion occurred between AS and SiC during laser micro-welding, and no new phases were generated. The results further confirmed that the bonding mechanism was a combination of metallurgical bonding dominated with minor mechanical interlocking (Fig. 6). Fig. 9b comparatively shows the XRD patterns in the 2θ from 33° to 39°. Obviously, the diffraction peaks shifted towards the right after welding, suggesting the lattice contraction and crystal surface spacing decreased [49].

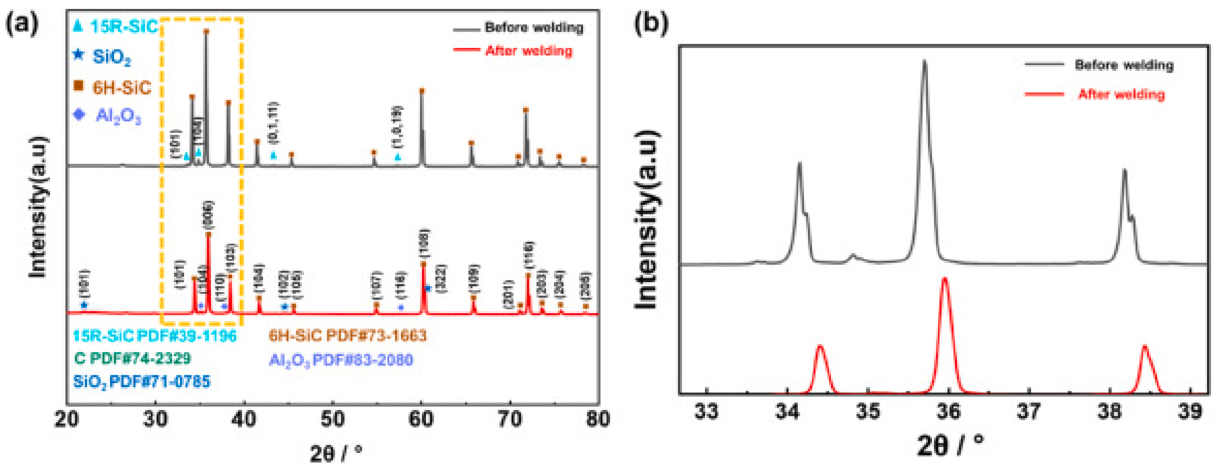

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Fig. 9. (a) XRD patterns at the SiC side of the fracture surface before and after welding; (b) partial enlarged view in (a).

3.5. Impermeability

Impermeability has been one of the key properties for the reliability of MEMS devices. More precisely, delicate MEMS devices were protected against environmental contaminants (*e.g.*, particulate matter and moisture ingress) through hermetic packaging architectures, ensuring operational stability under prolonged service conditions. These devices will fail if water and moisture, *etc.*, enter the functional components. Hence, impermeability testing is of importance.

The red ink immersion test was used to test the impermeability of the weld, as shown in Fig. 10a [50]. Fig. 10b-d shows the red ink impermeability test results of different scanning paths under optimal parameters. After soaking in a Petri dish for 2 h, the leakage rate for concentric circles was 8.14%, and that for grids was 16.34%. In terms of the filling grid inside circle, no leakage occurred within 2 h (Fig. 10d). Surprisingly, it lasted to 168 h without any leakage trace in the micro-welds (Fig. 10d3). Subsequently, the micro-welds were submerged to a depth of 1 m using a measuring cylinder for 30 min (Fig. 10e). Despite the application of high water pressure, no penetration of the red ink into the weld zone was observed. These results show that the welded samples maintain effective impermeability not only under conventional conditions but also in underwater environments, thereby satisfying the IPX7 waterproof standard (IEC 60529:2013, IDT) for enclosure packaging in practical applications. The results further demonstrated the feasibility of high-strength and impermeable AS/SiC joints fabrication *via* ultrafast laser micro-welding.

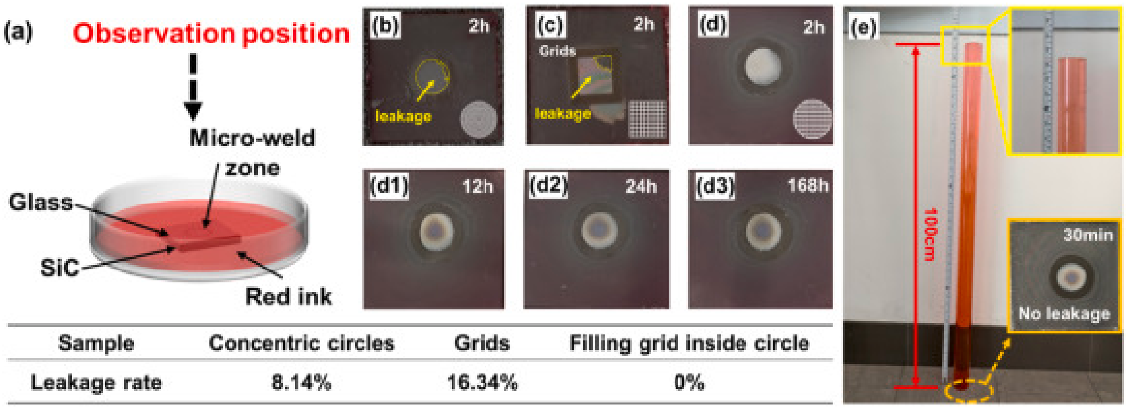

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Fig. 10. Red ink impermeability testing: (a) schematic diagram of red ink impermeability tests; (b) 2h concentric circles; (c) 2h grids; (d) 2h Filling grid inside circle; (d1)12h; (d2) 24h; (d3) 168h; (e) 1m deep red ink waterproof test.

3.6. Temperature field simulation

Fig. 11 (a) and (c) detail the temperature profiles at the starting points of laser scanning on the AS side (0, 0, 1) and on the SiC side (0, 0, −1) during the first pulse laser irradiation. Fig. 11a shows that the temperature at the AS side first increased to 2332.33K and then gradually decreased to 1072.08K. At the SiC side, the temperature rose sharply to 14652.05K and then gradually decreased to 976.95K (Fig. 10c). The softening temperature of AS was 1586K, while the melting point of SiC was 3393K, as indicated in Fig. 10. SiC does not melt under atmospheric pressure. When the temperature reached the melting point, the chemical bonds within the SiC lattice absorbed sufficient energy to fracture, resulting in thermal decomposition [51]. The significant disparity in the initial temperature was primarily attributed to the distinct mechanisms of light absorption. To be specific, the AS absorbed the laser energy by a non-linear absorption process, whereas it changed to a linear absorption process in the case of the SiC substrate [40,41]. Fig. 11b and (d) depict the temperature profiles at the laser scanning endpoint on the AS side (150, 0, 1) and the SiC side (150, 0,-1) during the first pulse period irradiation. At the AS side, the temperature was almost constant at 293.15K, which was close to room temperature (Fig. 11b). Differently, the temperature initially rose to 600.22K and then gradually decreased to 581 K at the SiC side. The variation in temperature profiles between AS and SiC was mainly attributed to the distinct heat transfer modes of the materials. The thermal conduction in both AS and SiC was predominantly governed by phonon transport [[52], [53]]. The nature of this transport is based on the thermal motion of protons, which behaves like a displacement wave propagating forward. In ideal conditions, this wave was elastic, and when quantized, it is called a “phonon” [54]. In contrast, the thermal conductivities of AS and SiC were different. The high thermal conductivity of SiC made it dissipate heat quickly, while the low thermal conductivity of AS dissipated heat slowly. Therefore, the temperature of SiC increased rapidly.

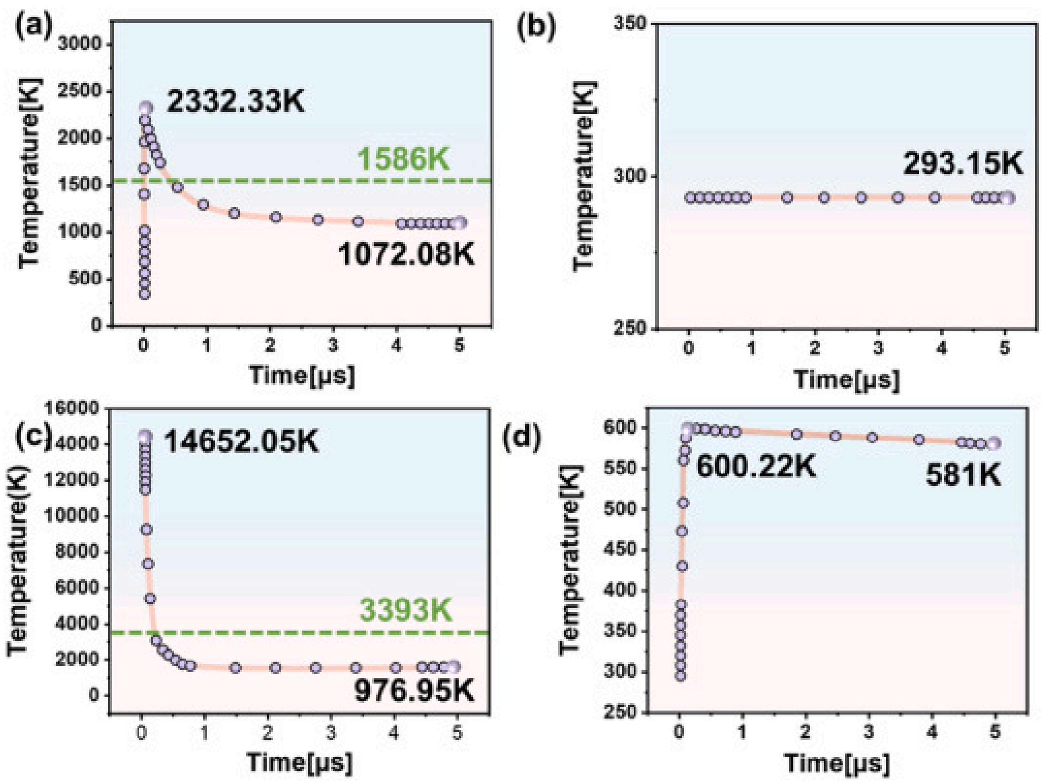

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Fig. 11. Temperature profiles at the probe point during the first pulse period of laser irradiation: (a) the AS side: at point (0, 0, 1); (b) the AS side: at point (150, 0, 1); (c) the SiC side: at point (0, 0, −1); (d) the SiC side: at point (150, 0, −1).

3.7. Joining mechanism

The formation mechanism of the joint was depicted schematically in Fig. 12. During the irradiation of AS and SiC with an ultrafast laser, it achieved sufficient focal intensity to induce nonlinear absorption [55]. The excitation of electrons in the materials was observed during this process, as shown in Fig. 12a. These photo-excited electrons then absorbed photon energy, which in turn initiated a cascade of ionization processes (Fig. 12b). Consequently, valence electrons acquired sufficient energy, enabling band-to-band transitions, resulting in a rapid increase in free electron density and a consequent sharp rose in temperature (Fig. 12c) [56]. After a series of reactions, the irradiated region underwent microstructural modification, forming a permanent micro-modified zone and micro-weld zone (Fig. 12d). When the laser penetrated the AS and reached the SiC surface, rapid energy absorption excited electrons in the material, triggering multiple ionization processes. Upon reaching critical ionization density, the liberated electrons and parent ions formed a high-temperature, high-density plasma *via* coulombic interactions (Fig. 12d1) [57,58]. The material temperature rapidly reached 14652.05 K (Fig. 11c). The high-temperature plasma erupted upward and upon interaction with the AS surface, induced rapid heating that elevated the local temperature to 2332.33 K, exceeding the AS softening temperature (Fig. 11a). Concurrently, the laser-generated AS plasma and SiC plasma interacted and mixed spatially, achieving energy exchange and coupling through processes such as photon absorption and electron collisions (*i.e.*, intra-plasma energy transfer mechanisms) (Fig. 12d) [59]. This interaction facilitated efficient energy transfer from the SiC to the AS interface, thereby enhancing atomic-scale bonding. The steep thermal gradient coupled with elemental heterogeneity within the molten pool triggered intense Marangoni convection and interdiffusion, while evaporation recoil pressure propelled molten SiC into the AS matrix [60]. In this process, the materials intermix and elemental diffusion occurred between SiC and AS (Fig. 12d3). Upon laser pulse termination, rapid solidification of the molten mixture layer formed a defect-free interfacial mixture layer at the AS/SiC interface (Fig. 12d4) [61]. Ultimately, a tight interfacial joint formed *via.* Dominated metallurgical bonding and minor mechanical interlocking. Thus, ultrafast laser processing offered a novel approach for AS/SiC heterogeneous integration.

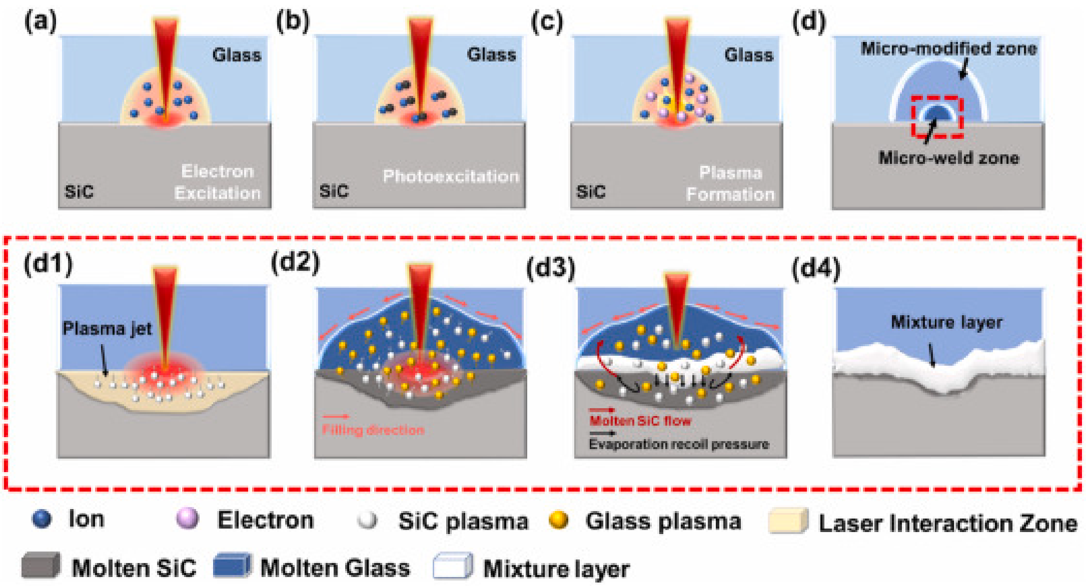

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Fig. 12. Schematic diagrams during ultrafast laser micro-welding of AS and SiC: (a–d) laser-material interaction; (d1) plasma jet formation; (d2) plasma diffusion and mixing; (d3) the molten flow; (d4) the mixture layer and final bonding formation.

4. Conclusion

This study proposes a new method with ultrafast laser micro-welding for the joining of AS and SiC. By using an orthogonal testing design, the optimal parameters were obtained. Macro- and Microstructural characterizations, joint shear strength, and permeability were performed. In addition, the joint formation mechanism was discussed in the aid of numerical simulation. The main conclusions were summarized as follows.

(1) Achieved high-quality bonding between AS and SiC, with a maximum mechanical strength of 33.1 MPa.

(2) The interfacial microstructure and fracture behavior analyses showed that intense material intermixing occurred in plasma and molten states, which facilitated element diffusion; whereas no new phase was

formed in the mixture layer at the interface. Concurrently, laser irradiation caused changes in the material's crystal structure.

(3) The AS/SiC joints with optimized parameters exhibited excellent impermeability, *viz.*, leakage free over 168 h by petri dish soaking and 30 min by measurement cylinder (1 m) soaking methods, which reached the IPX7 waterproof standard (IEC 60529:2013, IDT).

(4) COMSOL simulations analysis indicated that ultrafast laser irradiation induced transient high temperatures and localized plasma eruptions on the SiC side, which drove the following melt mixing and elemental diffusion between AS and SiC. Hence, it demonstrated a metallurgical dominated joining mechanism.

CRediT authorship contribution statement

Ming Wu: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Jin Yang:** Writing – review & editing, Resources, Funding acquisition. **Tao Zhang:** Validation, Project administration. **Yixuan Zhao:** Resources, Project administration. **Xiang Zhang:** Supervision, Investigation. **Qing Jiang:** Software, Methodology. **Jianian Tian:** Validation, Conceptualization. **Meng Yang:** Supervision, Methodology. **Tao Zhang:** Visualization, Investigation. **Peng Li:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

No data was used for the research described in the article.

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